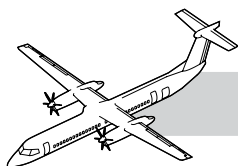


CHAPTER 55

STABILIZERS

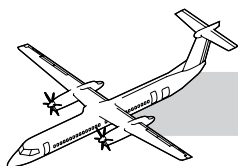
CONTENTS

	Page
55-00-00 INTRODUCTION	55-1
GENERAL	55-2
55-31-00 Vertical Stabilizer Access Panels	55-3
55-32-00 Vertical Stabilizer Fairings	55-4
COMPONENT DESCRIPTION	55-7
Horizontal Stabilizer	55-7
The Horizontal/Vertical Stabilizer Joint	55-7
Horizontal Stabilizer Main Structure	55-9
55-13-00 Horizontal Stabilizer Leading Edge	55-11
55-11-00 Horizontal Stabilizer Access Panels	55-12
55-14-00 Horizontal Stabilizer Wing Tip	55-13
Elevator Structure	55-14
55-21-00 Elevator Access Panels	55-15
55-22-00 Elevator Fairings	55-17
Rudder Structure	55-18
55-41-00-992-801 Rudder Access Panels	55-19
55-00-00 SPECIAL TOOL & TEST EQUIPMENT	55-20



ILLUSTRATIONS

Figure	Title	Page
55-1	Vertical Stabilizer Structure	55-2
55-2	Vertical Stabilizer Access Panels - Component Location	55-3
55-3	Vertical Stabilizer Fairing (Bullet Fairing) - Removal/Installation.....	55-4
55-4	Vertical Stabilizer Leading Edge - Removal/Installation	55-5
55-5	Horizontal Stabilizer.....	55-6
55-6	Horizontal Stabilizer Main Structure	55-8
55-7	Horizontal Stabilizer Leading Edge - Removal/Installation.....	55-10
55-8	Horizontal Stabilizer Access Panels - Component Location	55-12
55-9	Horizontal Stabilizer Wing Tip - Removal/Installation	55-13
55-10	Elevator Structure.....	55-14
55-11	Elevator Access Panels - Component Location Index (1 of 2)	55-15
55-12	Elevator Access Panels - Component Location Index (2 of 2)	55-16
55-13	Elevator Fairing - Removal/Installation.....	55-17
55-14	Rudder Structure.....	55-18
55-15	Rudder Access Panels - Component Location	55-19



CHAPTER 55 STABILIZERS



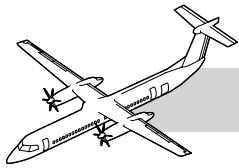
55-00-00 INTRODUCTION

The stabilizers give stable attitude around the lateral (pitch) and vertical (yaw) axis of the aircraft.

The stabilizer consists of four main sections as follows:

- Vertical stabilizer
- Horizontal stabilizer
- Elevator
- Rudder.

The basic structure is a conventional all metal construction utilizing high strength aluminium alloys (2024 and 7075) frames, stringers and chemical milled skin.



GENERAL

Refer to Figure 55-1. Vertical Stabilizer Structure.

The vertical stabilizer is an airfoil that stabilizes movement of the aircraft around the vertical (yaw) axis.

The vertical stabilizer is part of the aft/tail fuselage.

The vertical stabilizer has components that follow:

- Spar
- Ribs
- Stiffeners
- Rudder hinge fittings
- Skin
- Access panels and covers
- Leading edges
- Bullet fairings.

The stabilizer has three vertical spars connected by horizontal ribs, and is covered with skin panels.

The vertical spar members are shaped at the lower end to form the integral main frames of the aft/tail fuselage.

The leading edge has pneumatic deicer boots installed.

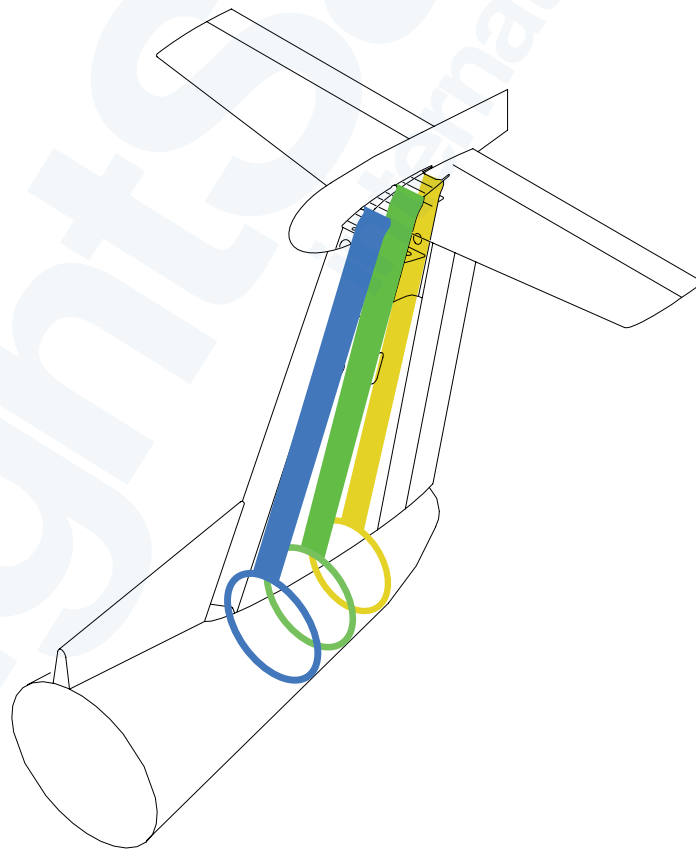
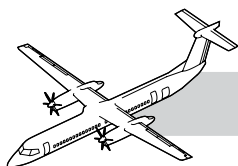


Figure 55-1. Vertical Stabilizer Structure



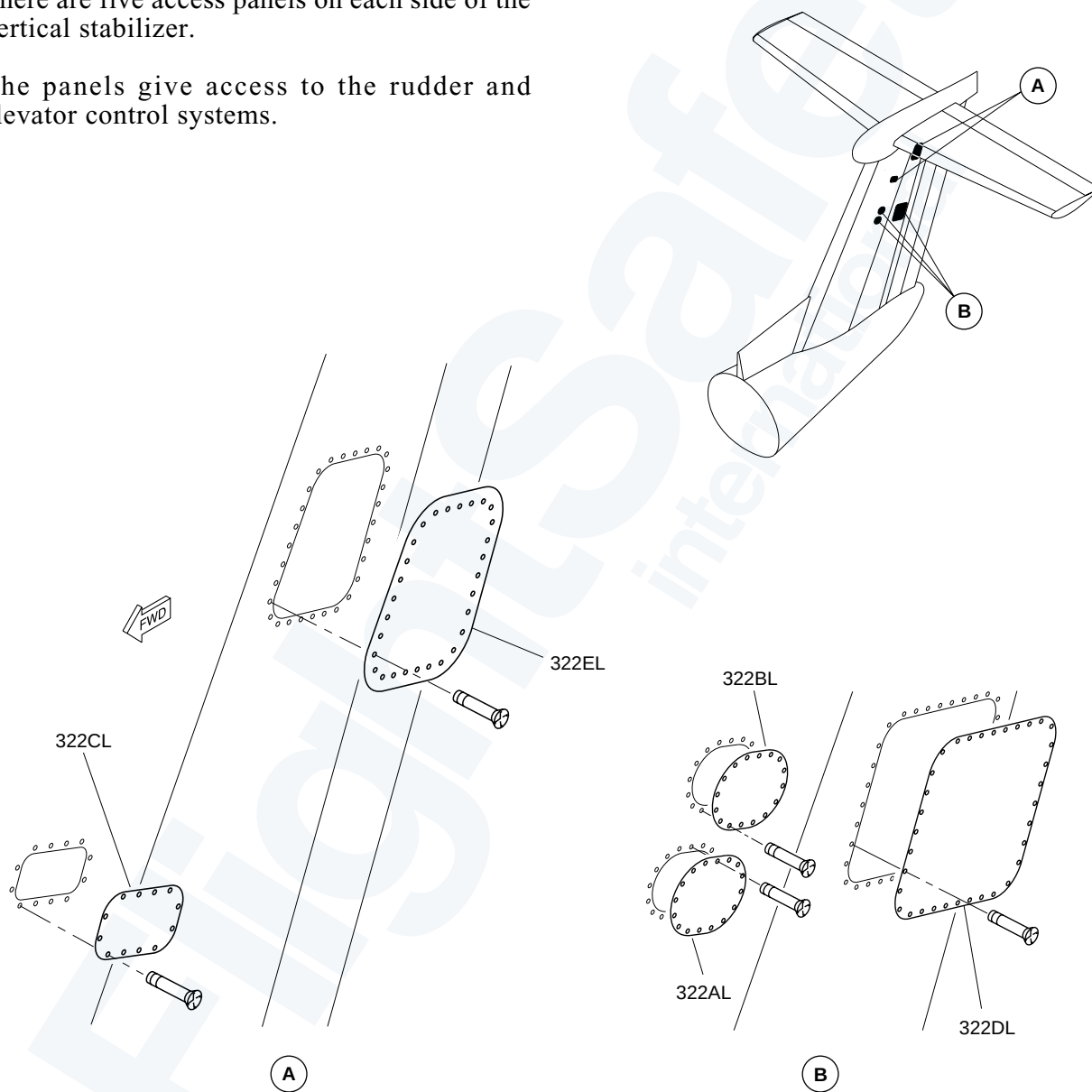
55-31-00 VERTICAL STABILIZER ACCESS PANELS

Refer to Figure 55-2. Vertical Stabilizer Access Panels - Component Location.

The vertical stabilizer has many panels to allow for maintenance access.

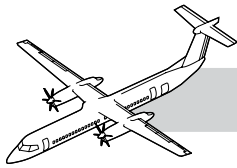
There are five access panels on each side of the vertical stabilizer.

The panels give access to the rudder and elevator control systems.



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Figure 55-2. Vertical Stabilizer Access Panels - Component Location



55-32-00 VERTICAL STABILIZER FAIRINGS

Refer to Figure 55-3. Vertical Stabilizer Fairing (Bullet Fairing) - Removal/Installation

The vertical stabilizer fairing is a combination of aerodynamic fairings. It starts at the top of the vertical stabilizer leading edge and continues aft to the rear of the horizontal stabilizer. The vertical stabilizer fairing is made from composite materials.

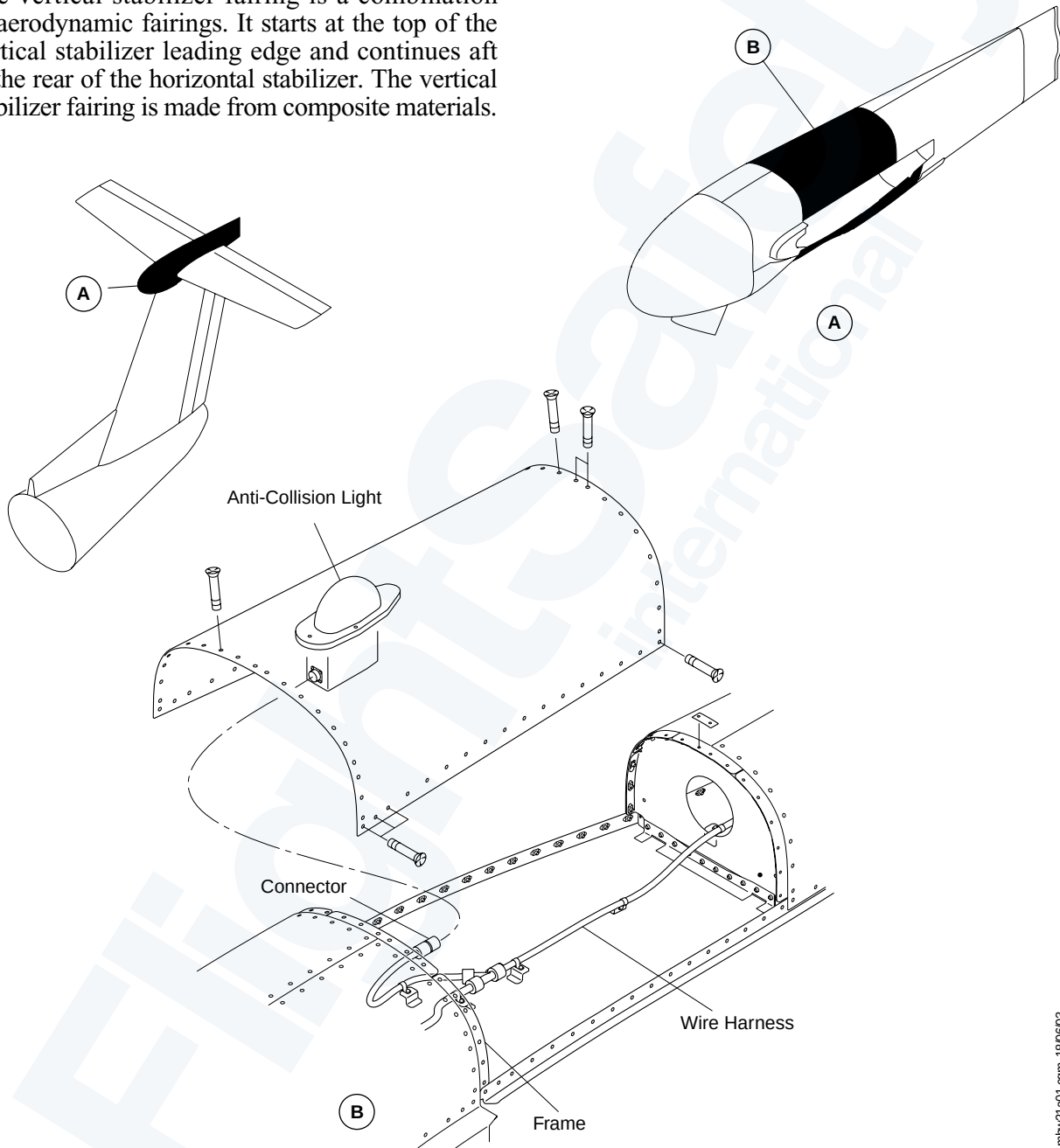
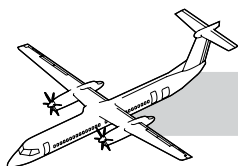


Figure 55-3. Vertical Stabilizer Fairing (Bullet Fairing) - Removal/Installation



Vertical Stabilizer Leading Edge

Refer to Figure 55-4. Vertical Stabilizer Leading Edge - Removal/Installation.

The vertical stabilizer leading edge is attached to the vertical stabilizer. It fairs with the aft dorsal fin at the lower end and with the vertical stabilizer fairing at the upper end. The vertical stabilizer leading edge may contain an H.F. antenna. The vertical stabilizer leading edge is made from composite materials.

Allowable Damage - Temporary Operations with Vertical Stabilizer Leading Edge Sealant Removed

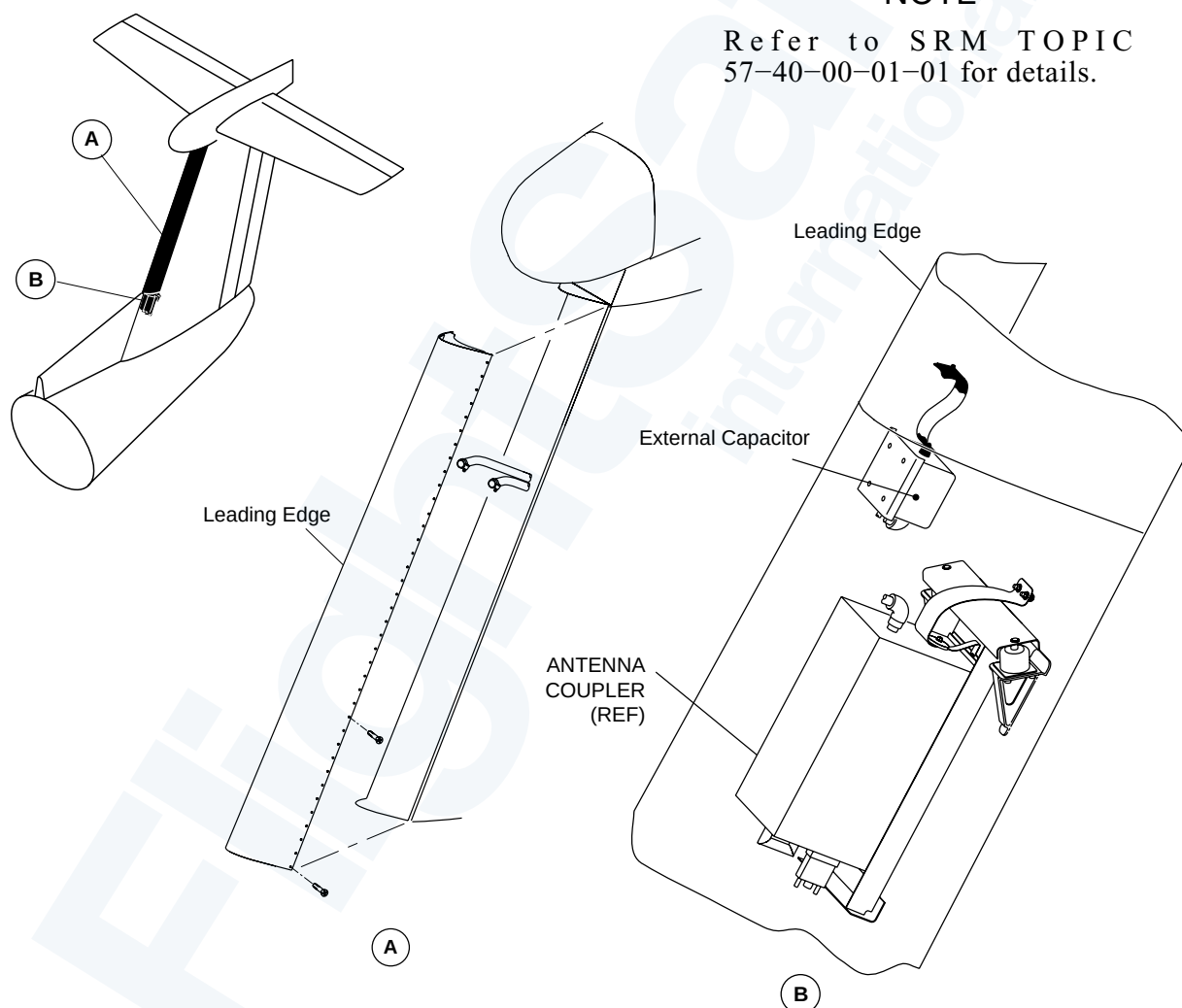
The leading edge sealant may be removed from up to one wing leading edge (per aircraft), one vertical stabilizer leading edge (per aircraft), and one horizontal stabilizer leading edge (per aircraft) simultaneously.

Continued flight operations is valid for 50 flight hours max.

At this time, the sealant must be installed.

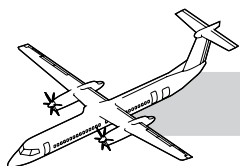
NOTE

Refer to SRM TOPIC 57-40-00-01-01 for details.



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Figure 55-4. Vertical Stabilizer Leading Edge - Removal/Installation



NOTE
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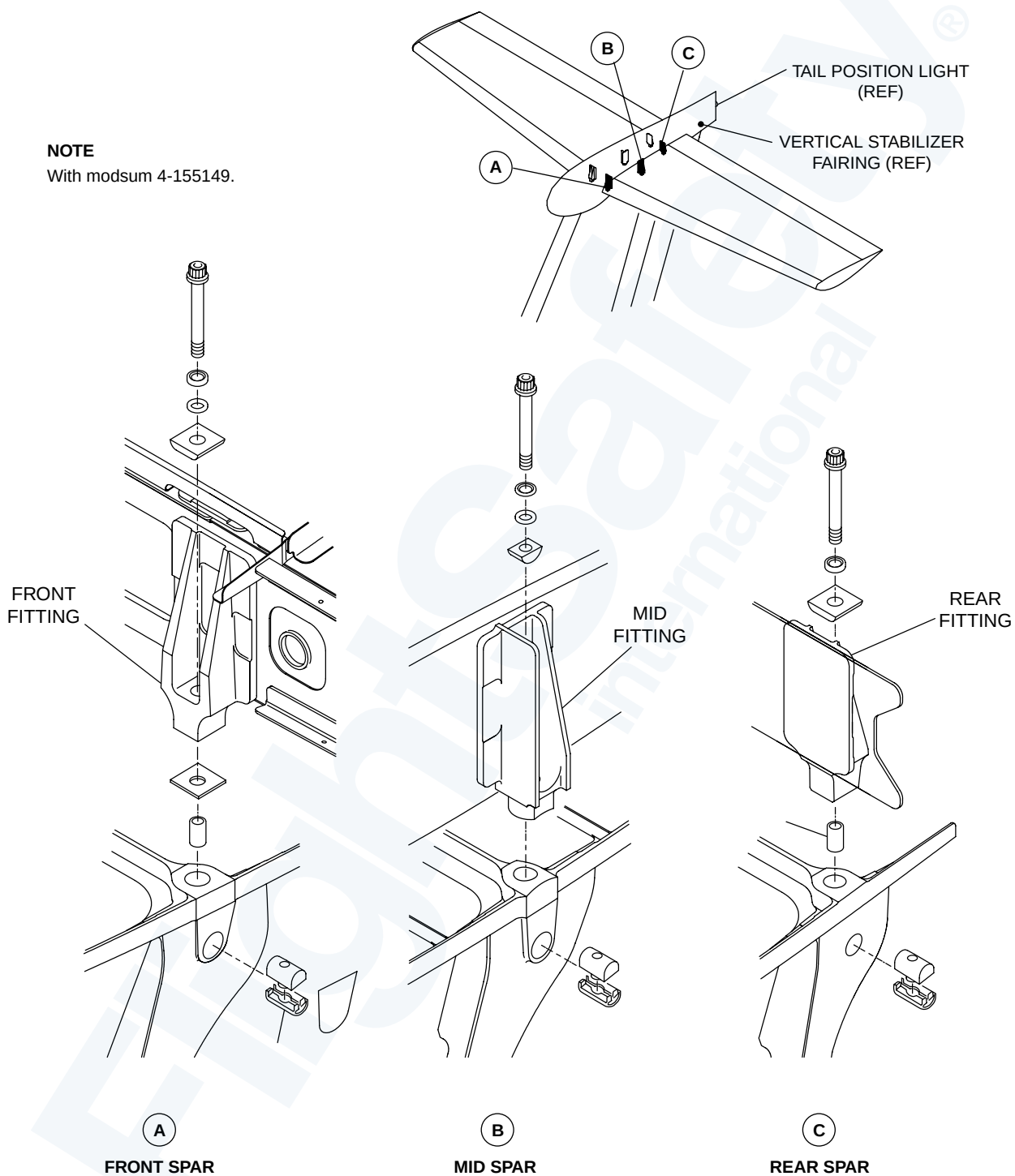
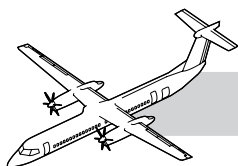


Figure 55-5. Horizontal Stabilizer



COMPONENT DESCRIPTION

NOTE

HORIZONTAL STABILIZER

Refer to Figure 55-5. Horizontal Stabilizer.

The horizontal stabilizer is attached to the top of the vertical stabilizer at six locations with bolts, retainers, and barrel nuts.

THE HORIZONTAL/VERTICAL STABILIZER JOINT

Refer to Figure 55-5. Horizontal Stabilizer.

The horizontal/vertical stabilizer joint is installed in the center box of the horizontal stabilizer and front spar of the vertical stabilizer. The joint between the vertical and horizontal stabilizer is covered by a removable composite construction aerodynamic fairing.

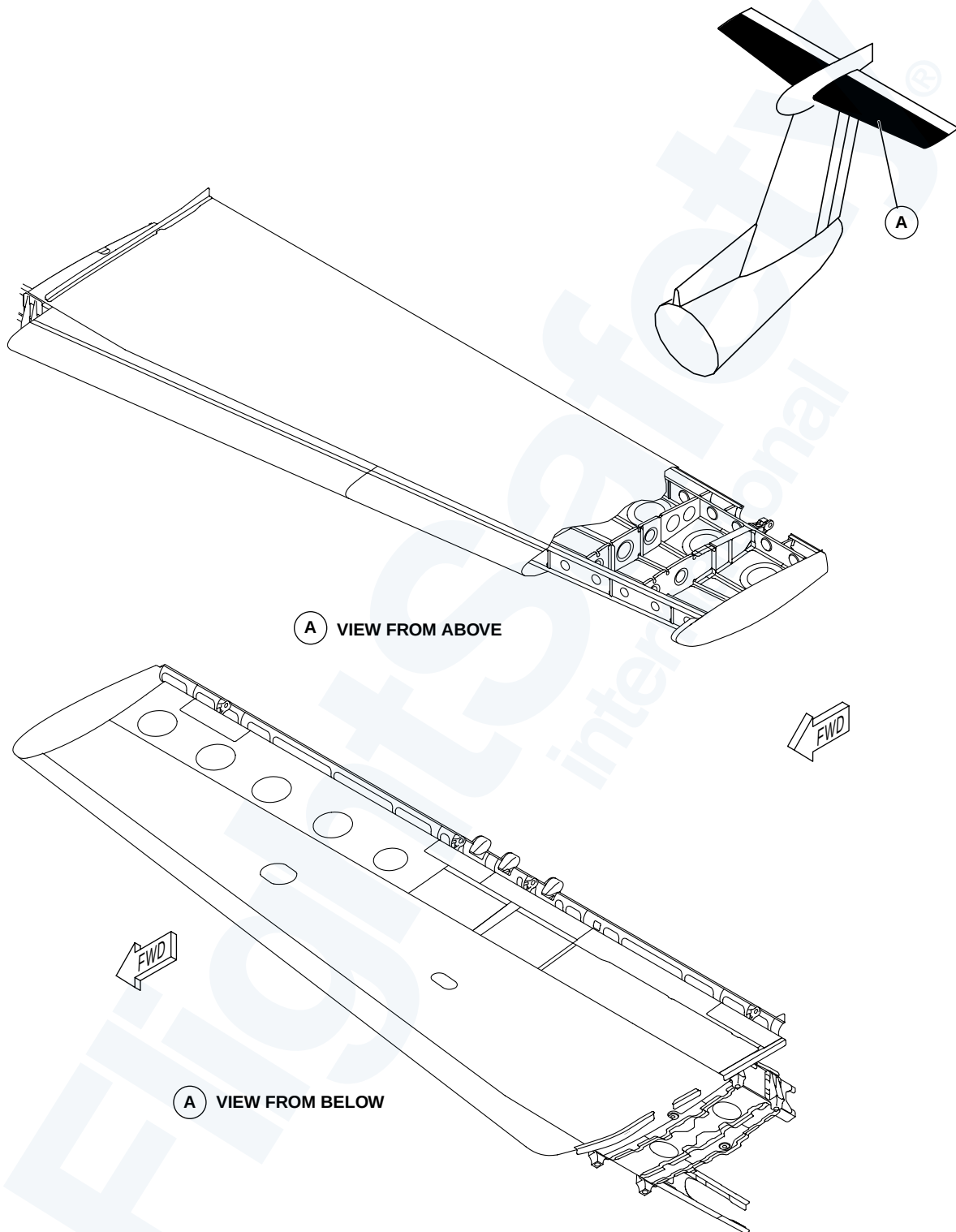
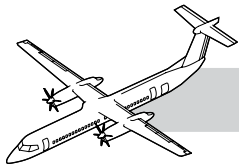
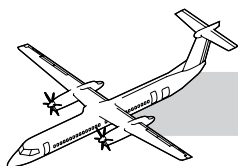


Figure 55-6. Horizontal Stabilizer Main Structure



HORIZONTAL STABILIZER MAIN STRUCTURE

NOTE

Introduction

The horizontal stabilizer is an airfoil that stabilizes movement of the aircraft around the lateral (pitch) axis.

Refer to Figure 55-6. Horizontal Stabilizer Main Structure.

The horizontal stabilizer has the components that follow:

- Stringers
- Spar
- Attached fittings
- Horizontal stabilizer skin
- Access panels
- Elevator hinge arms
- Leading edges
- Tip Fairing.

The fixed incidence horizontal stabilizer consists of a front, mid and rear spar connected by profile ribs and covered with skin panels stiffened by riveted stringers.

Each of the three spars has two fittings for attachment of the horizontal stabilizer at the upper end of the vertical stabilizer.

Hinge arms for the elevators are attached to the rear spar and adjacent ribs.

Leading edges are of a composite construction in two sections per side, fitted with locators to prevent incorrect installation, and are equipped with pneumatic deicer boots. The fairing is the secondary structure.

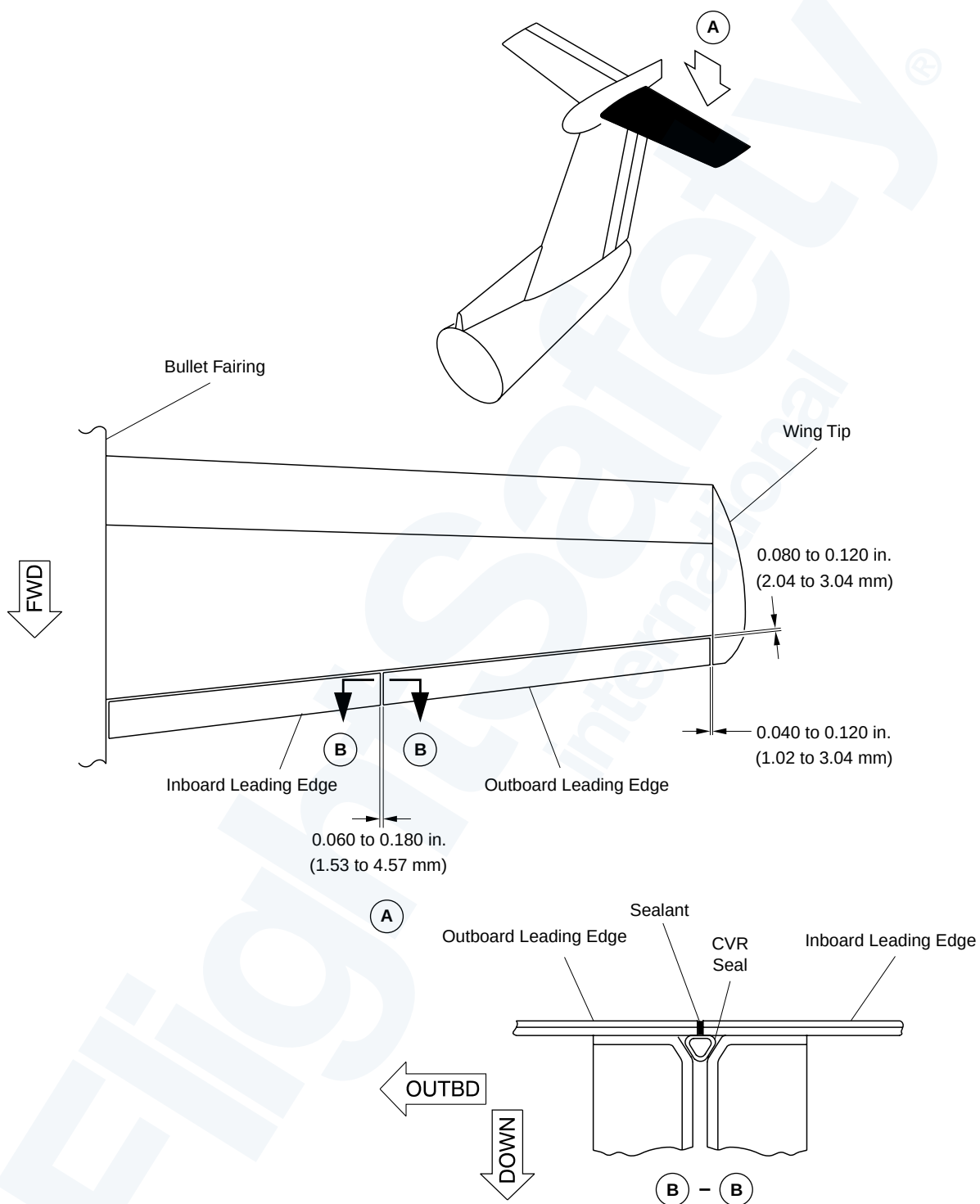
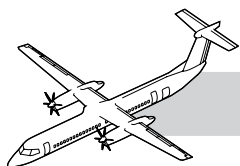
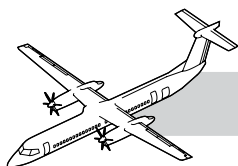


Figure 55-7. Horizontal Stabilizer Leading Edge - Removal/Installation



55-13-00 HORIZONTAL STABILIZER LEADING EDGE

Refer to Figure 55-7. Horizontal Stabilizer Leading Edge - Removal/Installation.

WARNING

HOLD THE LEADING EDGE DURING THE PROCEDURE. IF YOU DO NOT DO THIS, THE LEADING EDGE CAN FALL AND CAUSE INJURIES TO PERSONS AND DAMAGE TO THE EQUIPMENT.

CAUTION

DO NOT USE SHARP OBJECTS, BRUSHES OR ABRASIVE PADS TO REMOVE THE PROTECTIVE LATEX LAYER. IF YOU DO THIS, YOU WILL CAUSE DAMAGE TO THE DE-ICING BOOTS.

NOTE

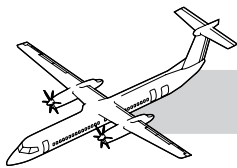
It is permitted to operate the aircraft with the leading edge sealant removed from up to one wing leading edge, and one vertical stabilizer leading edge, and one horizontal stabilizer leading edge simultaneously from an aircraft.

NOTE

The aircraft can be operated until it is convenient to apply sealant, or within 50 flight hours, whichever ever occurs first.

NOTE

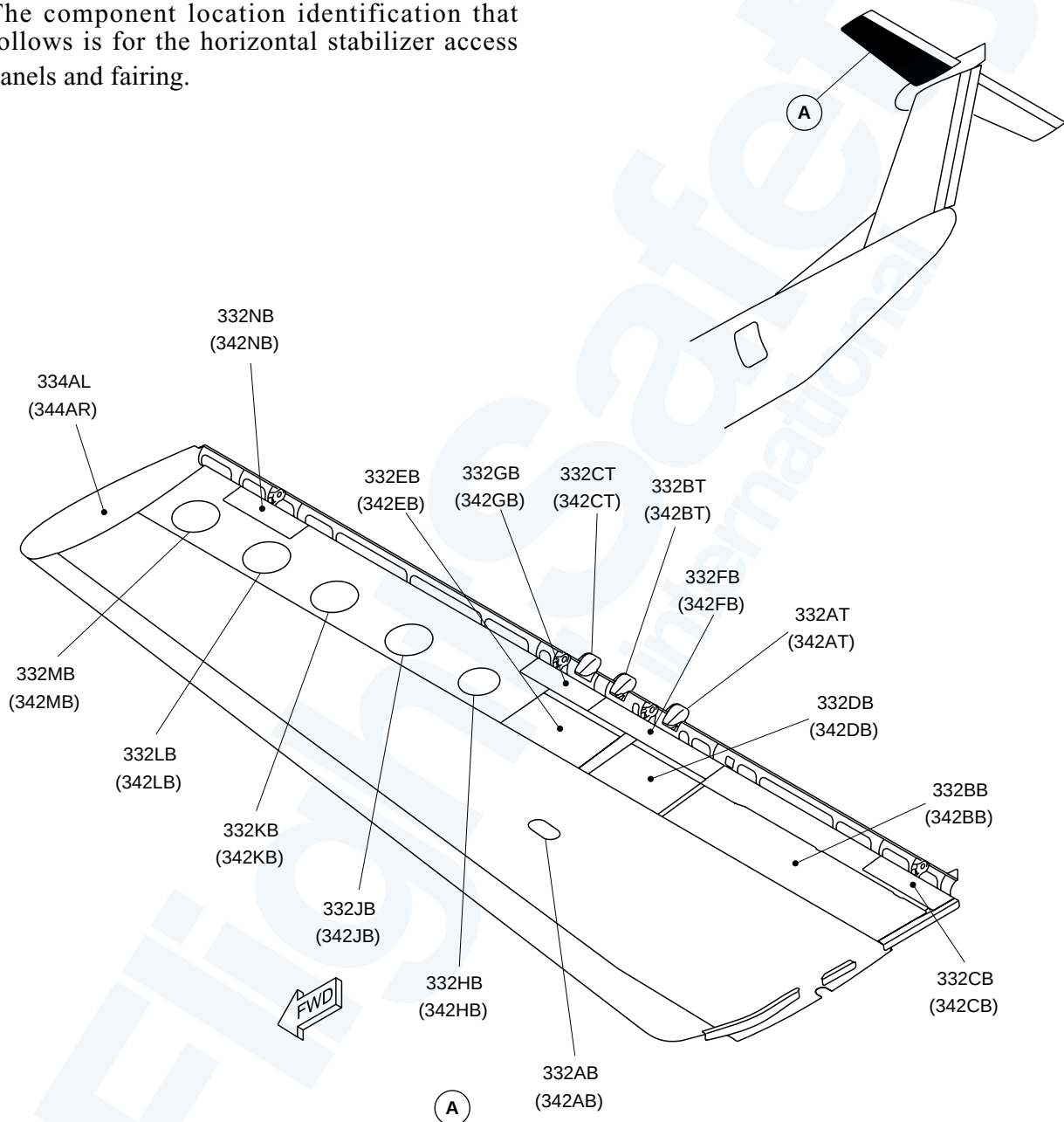
If the aircraft has been operated without the sealant, the leading edge must be removed and the related wing/stabilizer are inspected and cleaned of moisture/deicing fluid before sealant can be applied.



55-11-00 HORIZONTAL STABILIZER ACCESS PANELS

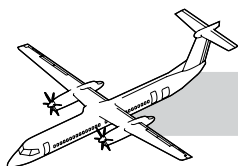
Refer to Figure 55-8. Horizontal Stabilizer Access Panels - Component Location.

The component location identification that follows is for the horizontal stabilizer access panels and fairing.



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Figure 55-8. Horizontal Stabilizer Access Panels - Component Location



55-14-00 HORIZONTAL STABILIZER WING TIP

Refer to Figure 55-9. Horizontal Stabilizer Wing Tip - Removal/Installation.

The wing tip provides an aerodynamic cover for the end rib of the horizontal stabilizer.

The wing tip is installed at the ends of the horizontal stabilizer.

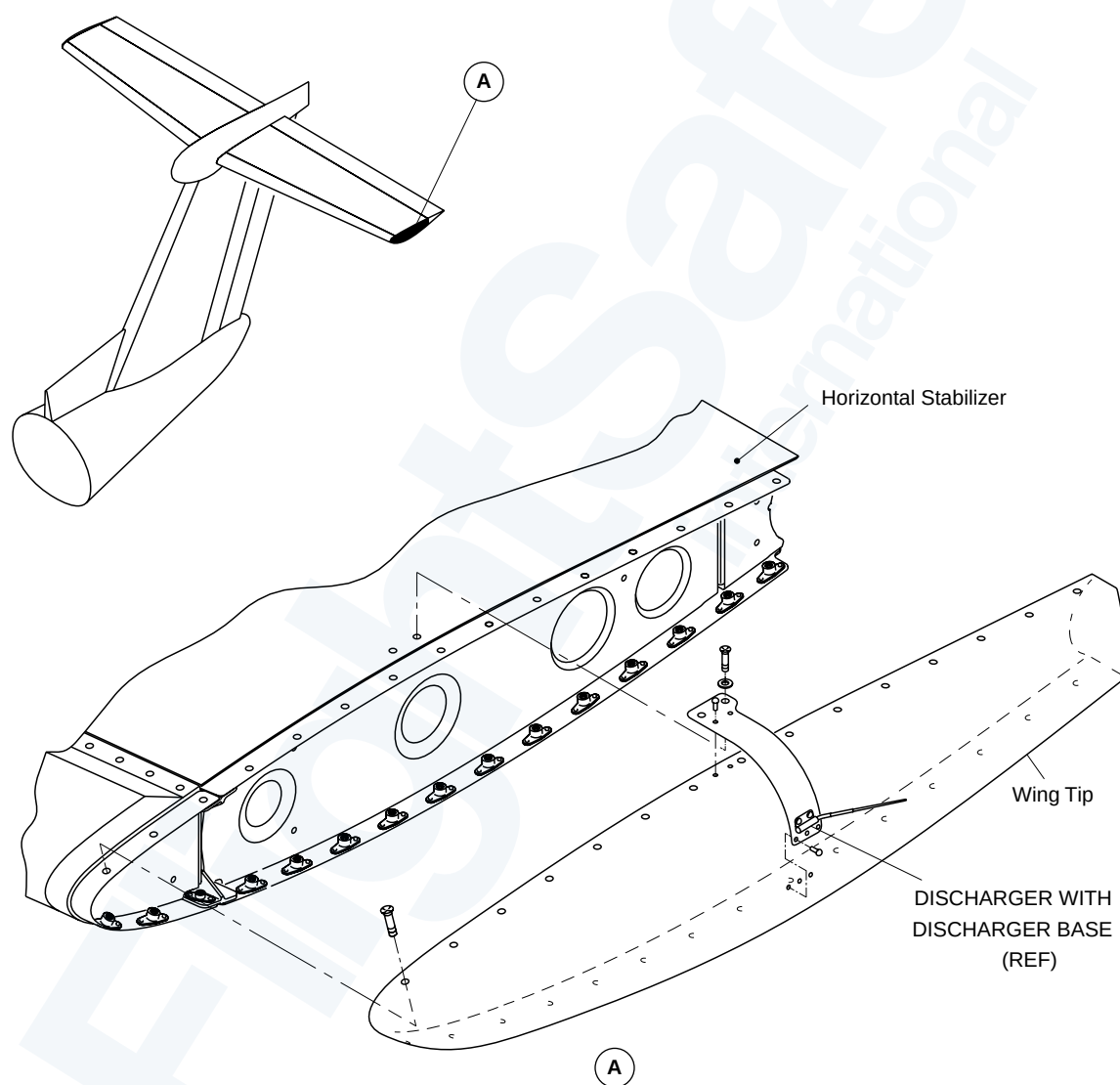
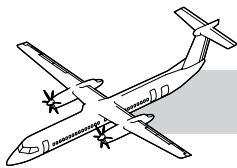


Figure 55-9. Horizontal Stabilizer Wing Tip - Removal/Installation



ELEVATOR STRUCTURE

Introduction

Refer to Figure 55-10. Elevator Structure.

The left and the right elevators are movable flight control surfaces that control the movement of the aircraft around the lateral axis.

The elevators have the components that follow:

- Spar
- Ribs
- Elevator hinge fittings
- Elevator skin
- Access panels
- Elevator fairing.

The left and right elevators consist of a main and rear spar interconnected by profile ribs and covered by skin panels.

The inboard half of the elevator also incorporates a front spar fitted with a leading edge sealing strip.

The elevators are moved directly by hydraulic power.

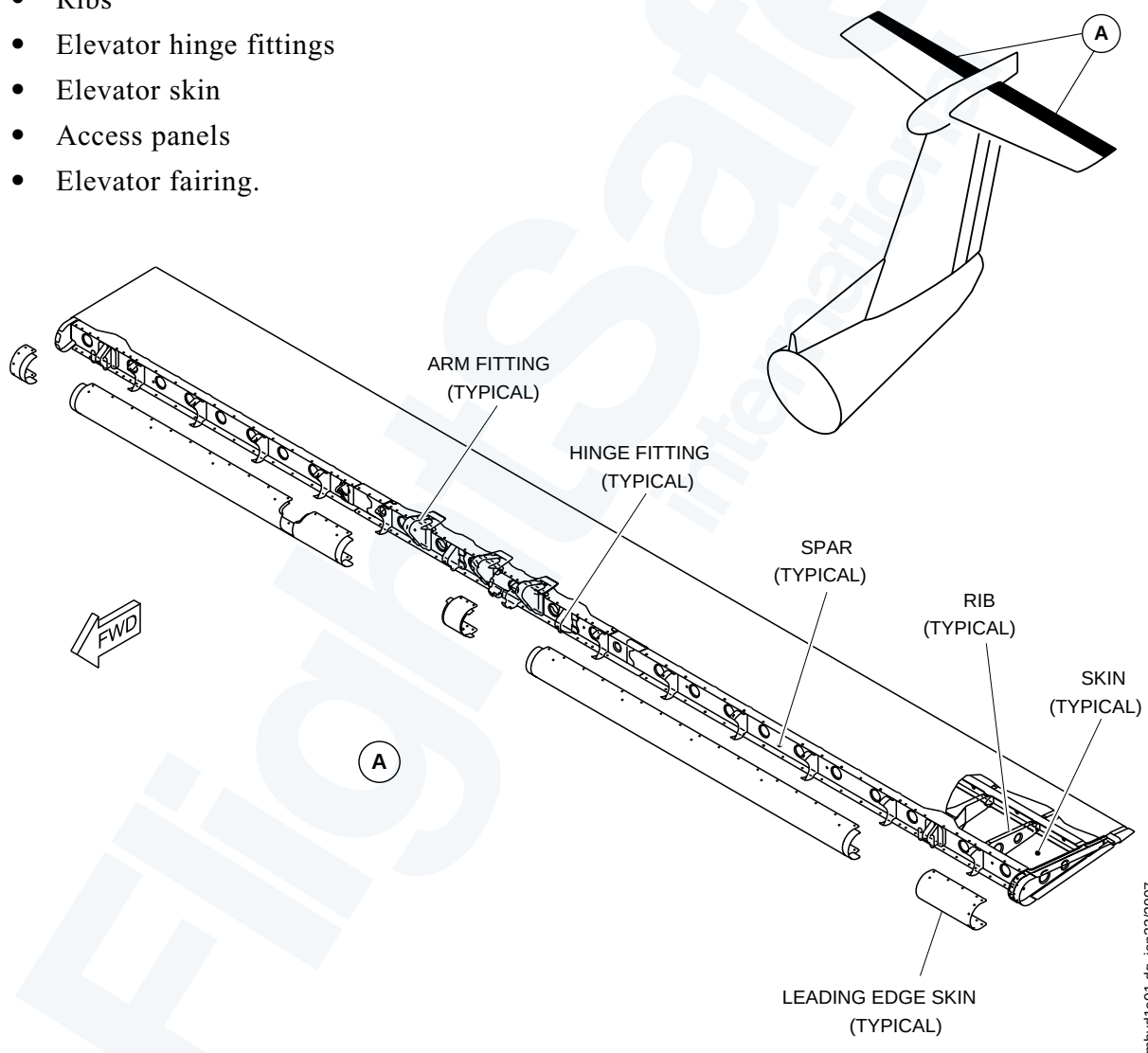
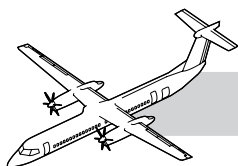


Figure 55-10. Elevator Structure

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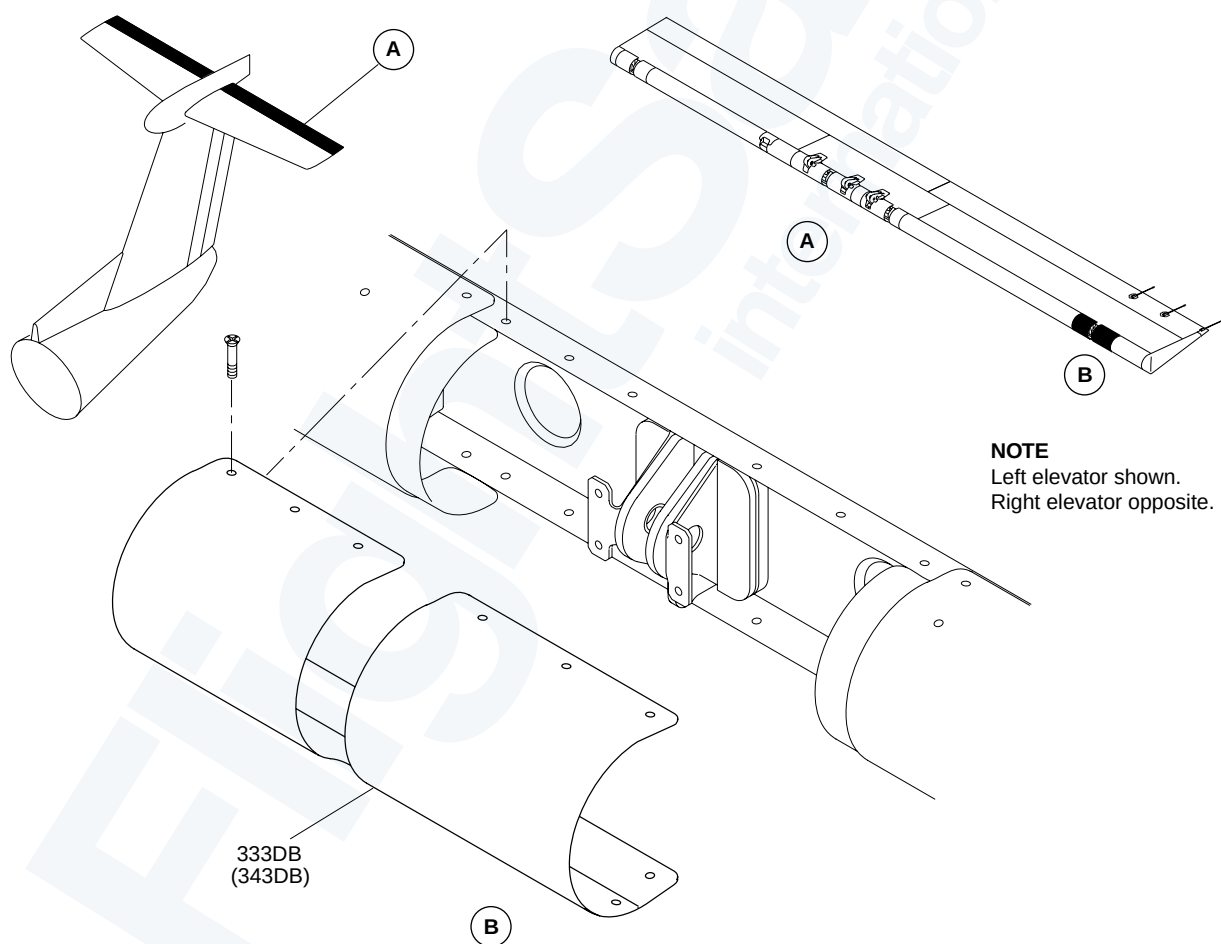
55-21-00 ELEVATOR ACCESS PANELS

Refer to:

- Figure 55-11. Elevator Access Panels - Component Location Index (1 of 2).
- Figure 55-12. Elevator Access Panels - Component Location Index (2 of 2).

The elevator access panels are located on the leading edge of the elevators.

These panels allow maintenance access to the hinge bolts for elevator removal.



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Figure 55-11. Elevator Access Panels - Component Location Index (1 of 2)

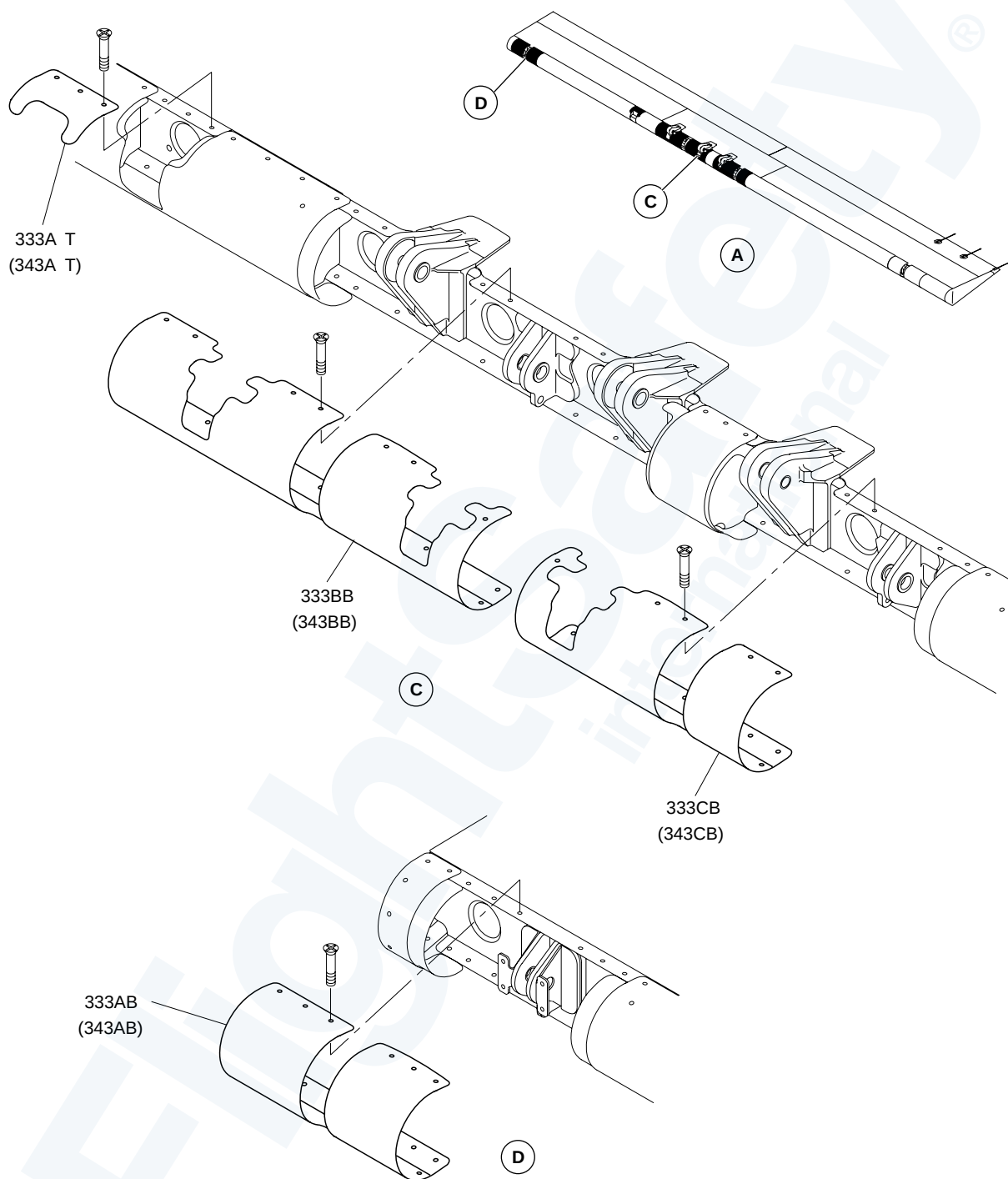
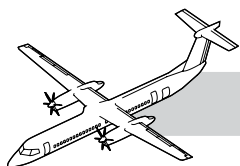
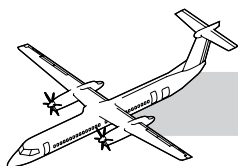


Figure 55-12. Elevator Access Panels - Component Location Index (2 of 2)

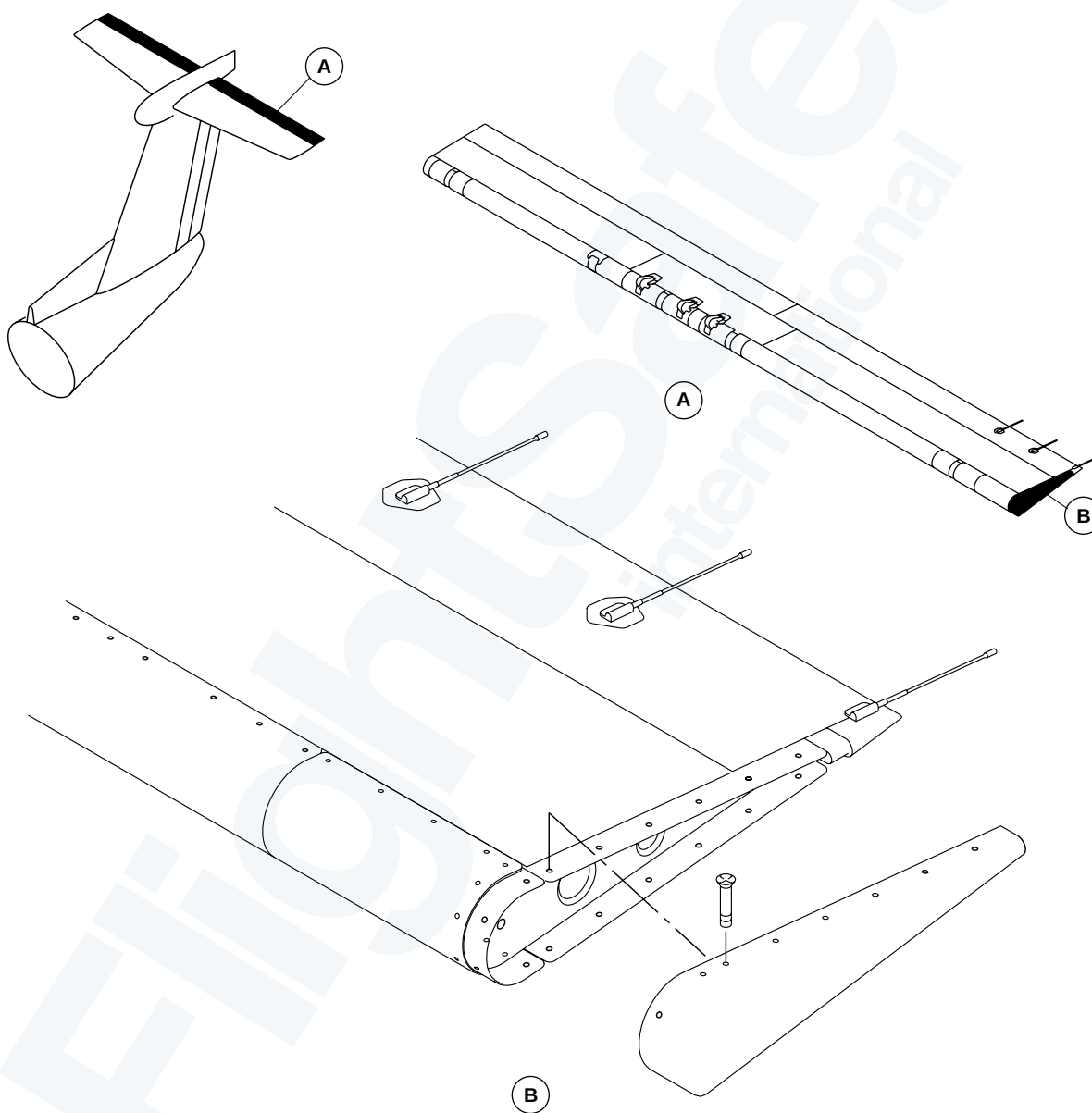
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55-22-00 ELEVATOR FAIRINGS

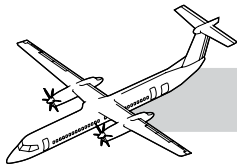
Refer to Figure 55-13. Elevator Fairing
- Removal/Installation.

The fairing is located on the outboard end of the elevator and forms an aerodynamic closure for the end rib.



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Figure 55-13. Elevator Fairing - Removal/Installation



RUDDER STRUCTURE

Introduction

Refer to Figure 55-14. Rudder Structure

The rudder is a moveable flight control surface that controls movement of the aircraft around the vertical axis.

The rudder has a fore rudder and a trailing rudder.

Both have the components that follow:

- Spar
- Ribs
- Hinge fittings
- Skin.

The fore rudder is hinged to the vertical stabilizer rear spar, and the trailing rudder is hinged to the trailing edge of the fore rudder.

The fore rudder is moved directly by hydraulic power. Mechanical linkage transmits fore rudder movement to the trailing rudder.

Both rudders are constructed of spars and profile ribs covered by skin panels.

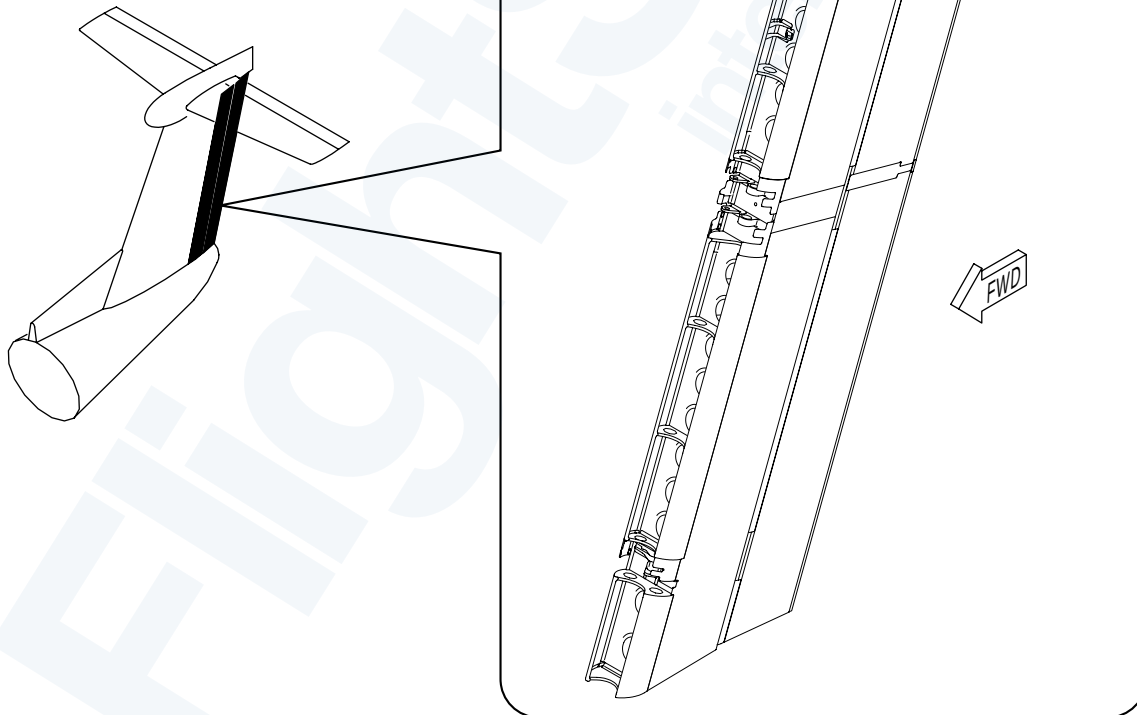
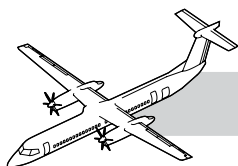


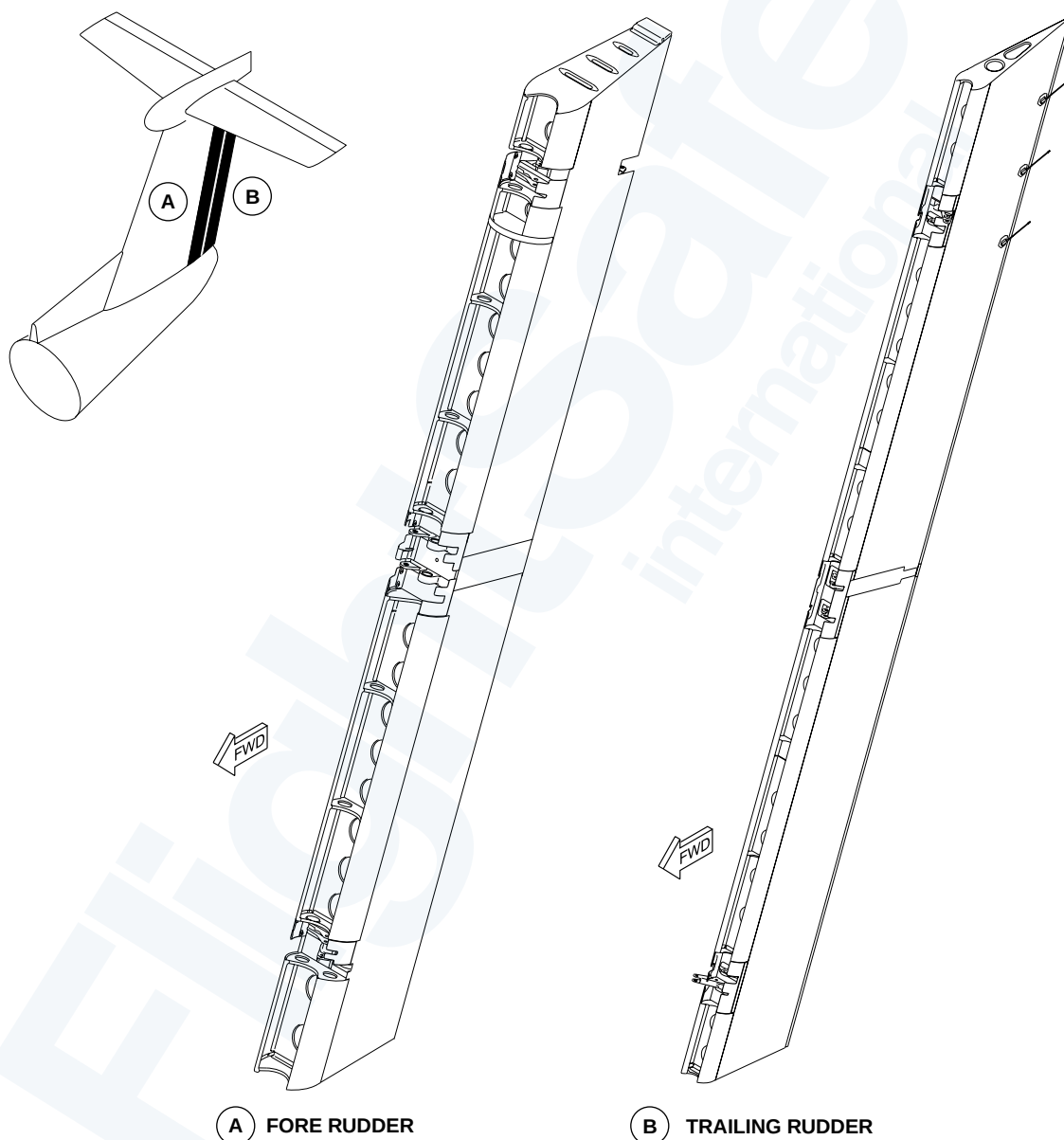
Figure 55-14. Rudder Structure



55-41-00-992-801 RUDDER ACCESS PANELS

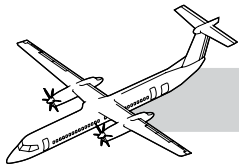
Refer to Figure 55-15. Rudder Access Panels - Component Location.

There are four access panels on the left and right side of each rudder leading edge. The panels give access to the rudder hinge and PCU attachments.



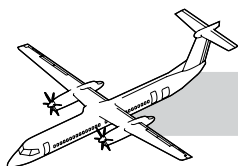
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Figure 55-15. Rudder Access Panels - Component Location



55-00-00 SPECIAL TOOL & TEST EQUIPMENT

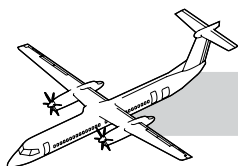
- GSB2700008 KIT - RIGGING PIN (SERIES 400)
- GSB2700008-03 Rigging Pin, 0.250 - 3.00 in (6.35 - 76.20 mm) L-Pin
- GSB2700008-05 Rigging Pin, 0.250 - 6.00 in (6.35 - 152.4 mm) L-Pin
- Commercially available Horizontal Stabilizer Lifting Tool
- GSB2000001 Borescope - 110 Volt, 60 Hz (if necessary)
- GSB2000015 Borescope - 220 Volt, 50 Hz (if necessary)



CHAPTER 56 WINDOWS

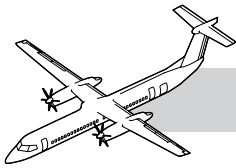
CONTENTS

	Page
56-00-00 INTRODUCTION	56-1
GENERAL	56-5
Training Information Point.....	56-5
Passenger Compartment Windows.....	56-9
Special Tooling	56-9
Safety Precautions.....	56-9



ILLUSTRATIONS

Figure	Title	Page
56-1	Windows.....	56-2
56-2	Flight Compartment Windshield	56-4
56-3	Flight Compartment Side Windows.....	56-6
56-4	Passenger Compartment and Type II - Emergency-Exit-Door Windows	56-8
56-5	Aft Passenger Door Window	56-10



CHAPTER 56 WINDOWS



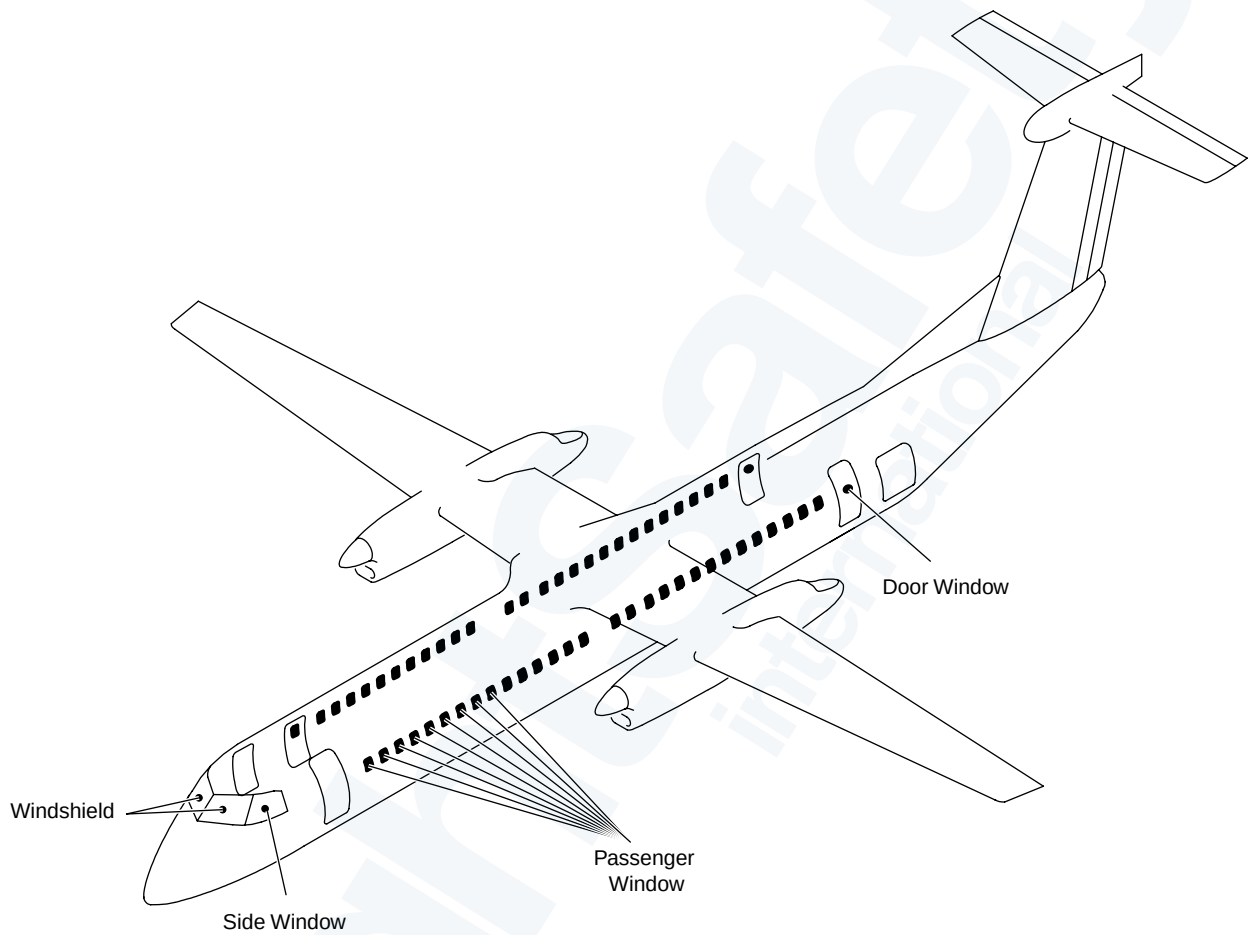
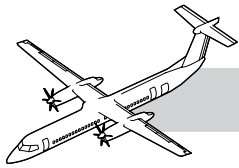
56-00-00 INTRODUCTION

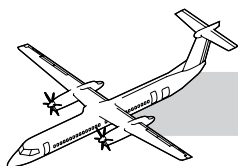
Refer to Figure 56-1. Windows

The Dash 8 Q400 Series aircraft has three classifications for windows that follow:

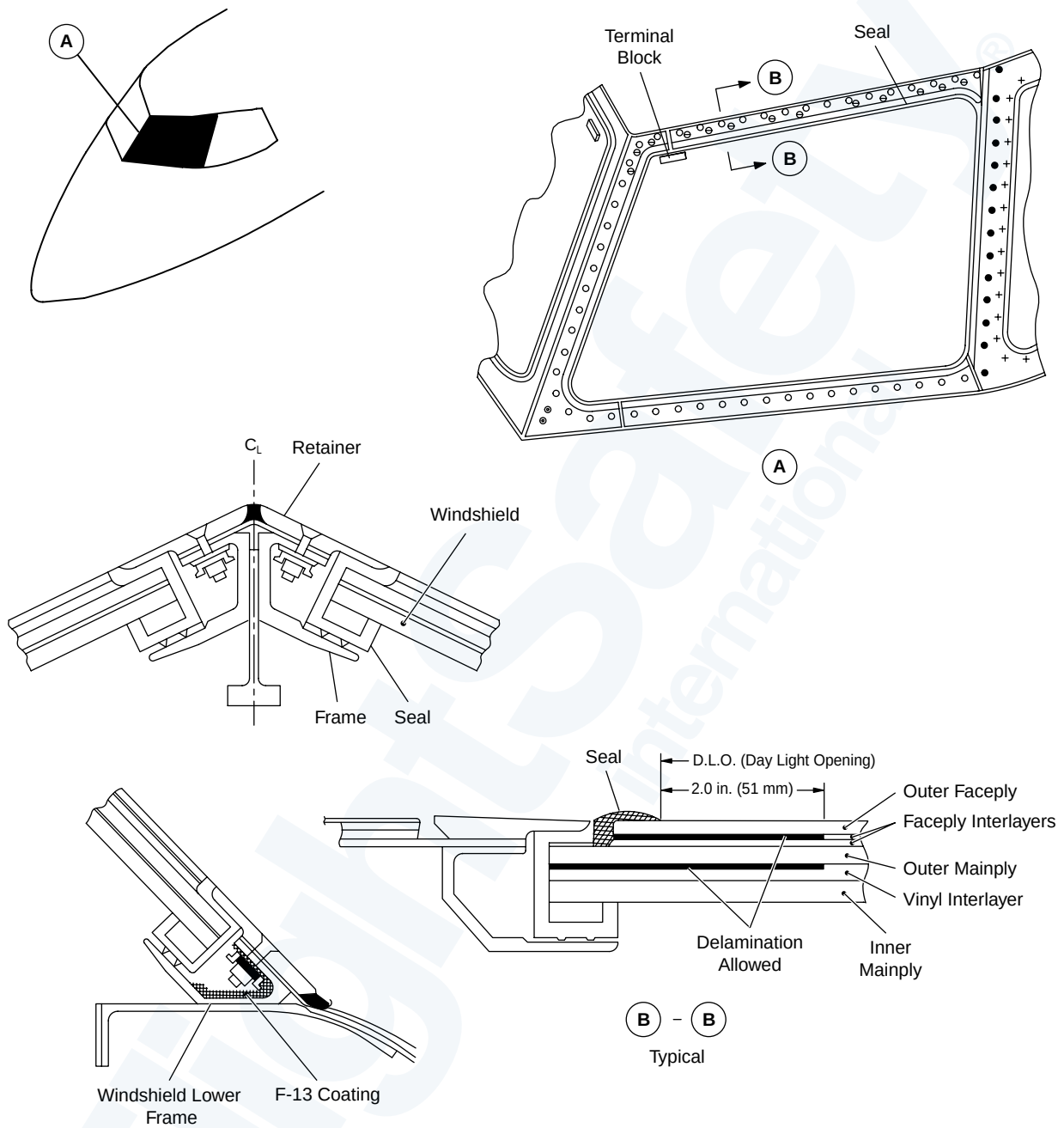
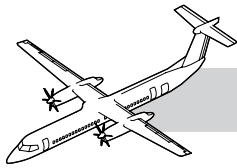
- Flight compartment windshields and side windows
- Passenger (compartment) cabin
- Door.

The flight compartment windows give the flight crew an adequate, unobstructed view outside the aircraft under all flight and weather conditions. The windows are structural elements which are able to withstand pressurization forces and impact from a bird strike. The passenger compartment windows give the passengers a view outside the aircraft. The windows are able to withstand pressurization forces and insulate the passengers from the extreme temperatures outside the aircraft.

**Figure 56-1. Windows**

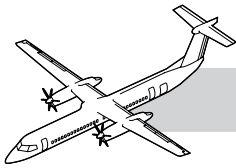


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**NOTE**

Total delamination not more than 2.0 inches (51mm) from Day Light Opening (DLO) of the seal.

Figure 56-2. Flight Compartment Windshield



GENERAL

Refer to Figure 56-2. Flight Compartment Windshield

The windshields are installed in the forward flight compartment. One is for the pilot; one for the co-pilot. The windshields are installed in an aluminium alloy window frame with four window retainers that are bolted to the frame. One retainer is also for the side window and is installed before the windshield is installed.

The windshields have panes that follow:

- Outer face ply
- Face ply interlayers
- Outer main ply
- Vinyl interlayer
- Inner main ply.

Each windshield has a thermally tempered glass outer face ply that is coated on the internal side with a conductive, metallic film. This supplies heating for anti-icing. The heating film covers most of the windshield area except for the forward section. Two urethane face ply interlayers are adjacent to the outer glass face ply.

The window also has two chemically strengthened glass main plies that are separated by a vinyl interlayer. A molded silicone rubber seal is bonded around the edge of the windshield for weather and pressure sealing. The two chemically strengthened glass main plies are the structural part of the windshield.

A terminal block and three temperature sensors are located near the top of the inboard side of the windows. The temperature sensors are located between the outer face and interlayer face plies.

The sensors give an interface between the windshields and the anti-icing system for window heating.

The windshield gives a satisfactory view of the aircraft for all flight and weather conditions. The windshields are structural parts that can withstand pressurization forces and an impact from a bird strike.

TRAINING INFORMATION POINT

The main types of damages that can occur to the windshields are as follows:

Scratches:

The three groups of scratches are:

- Faint scratches that can be seen but can not be felt with your fingernails
- Minor scratches that can be seen and felt with the fingernails < 0.004" (1.0mm)
- Major > 0.004" (1.0mm).

For details and depth measurement details please refer to the Aircraft Maintenance Manual.

Delaminating:

It is the separation of two or more layers of a laminated window panel. It usually occurs at the edges and/or at the corners of the window panel.

Cracks:

They are opening through all of the thickness of the window panels.

Bubbles:

Bubbles are trapped air within the interlayer usually circular in appearance.

Arcing:

It is an electrical discharge across a discontinuity in a wire, bus bar or electrical conductor.

Maintenance Practices/Ground Handling Requirements

As the inner pane of these windows have very little tolerance for damage, it is therefore very important to keep anything that can cause damage off the glare shield and away from the surface of the glass. This includes the journey logbook, pens, pencils, watches and other jewelry, and tools.

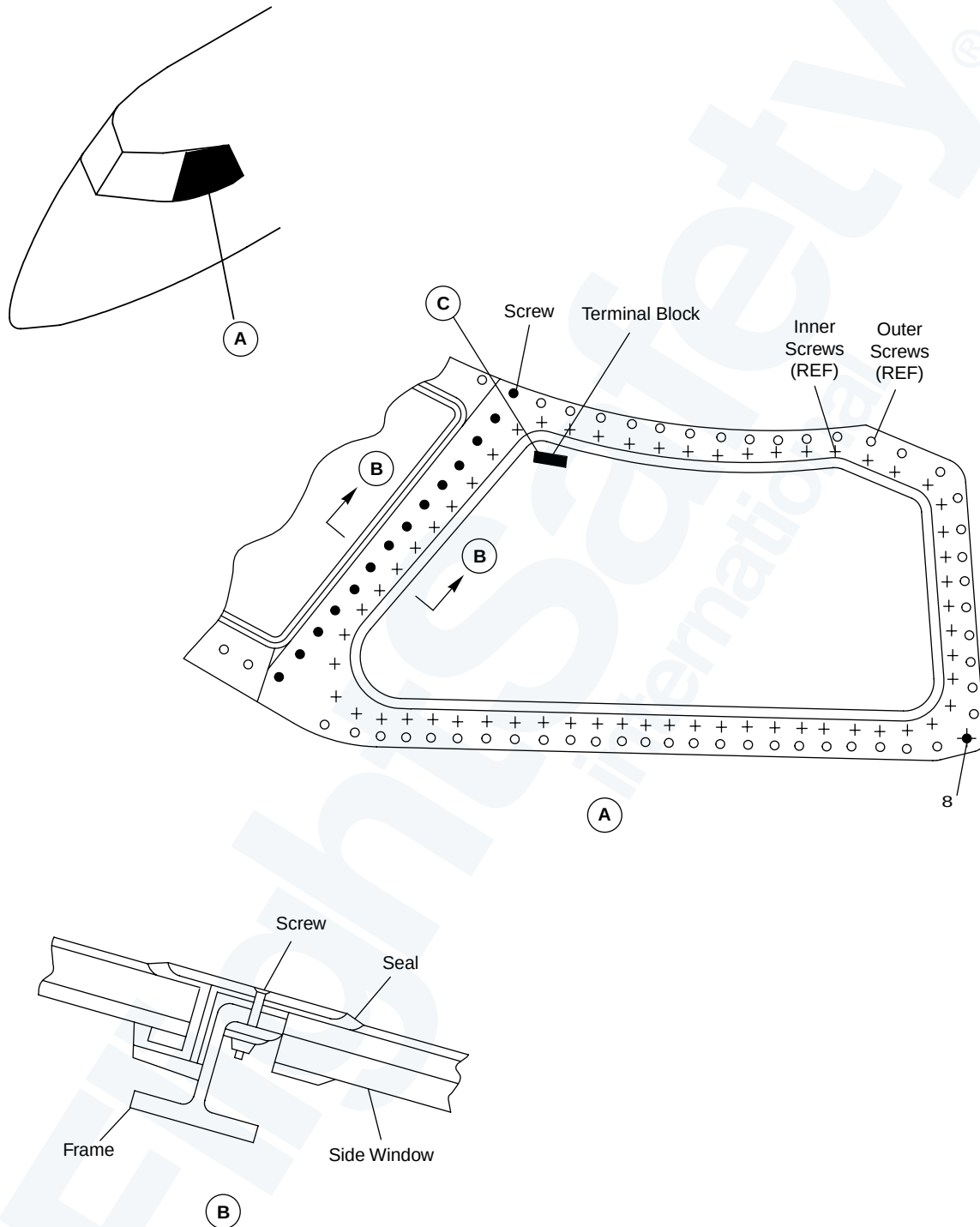
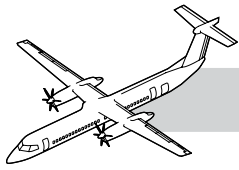
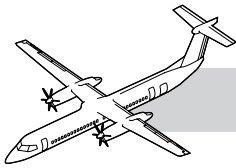


Figure 56-3. Flight Compartment Side Windows



Refer to Figure 56-3. Flight Compartment Side Windows

NOTE

The side windows are installed immediately aft of the pilot's and co-pilot's windshield. They are installed in an aluminium alloy window frame with a retainer that is bolted to the window and to the frame.

The pilot's and co-pilot's side windows are similar.

The pilot's side window has an outer, non-structural, acrylic face ply with a structural stretched acrylic main ply. The forward part of the face ply is coated on the internal side with a conductive, metallic film for anti-icing. There is a molded polysulfide weather seal and a rubber pressure seal around the edge of the window. A retainer is attached to the window with inner screws.

NOTE

The inner screws of the retainer are torqued by the manufacturer of the window and may be tightened only if air leakage occurs. If an inner screw is sheared, it must be replaced immediately. If more than five screws are sheared during one flight, the side window must be replaced.

A terminal block and three temperature sensors are located near the top of the inboard side of the pilot's side window.

They give an interface between the pilot's side windows and the anti-icing system for window heating.

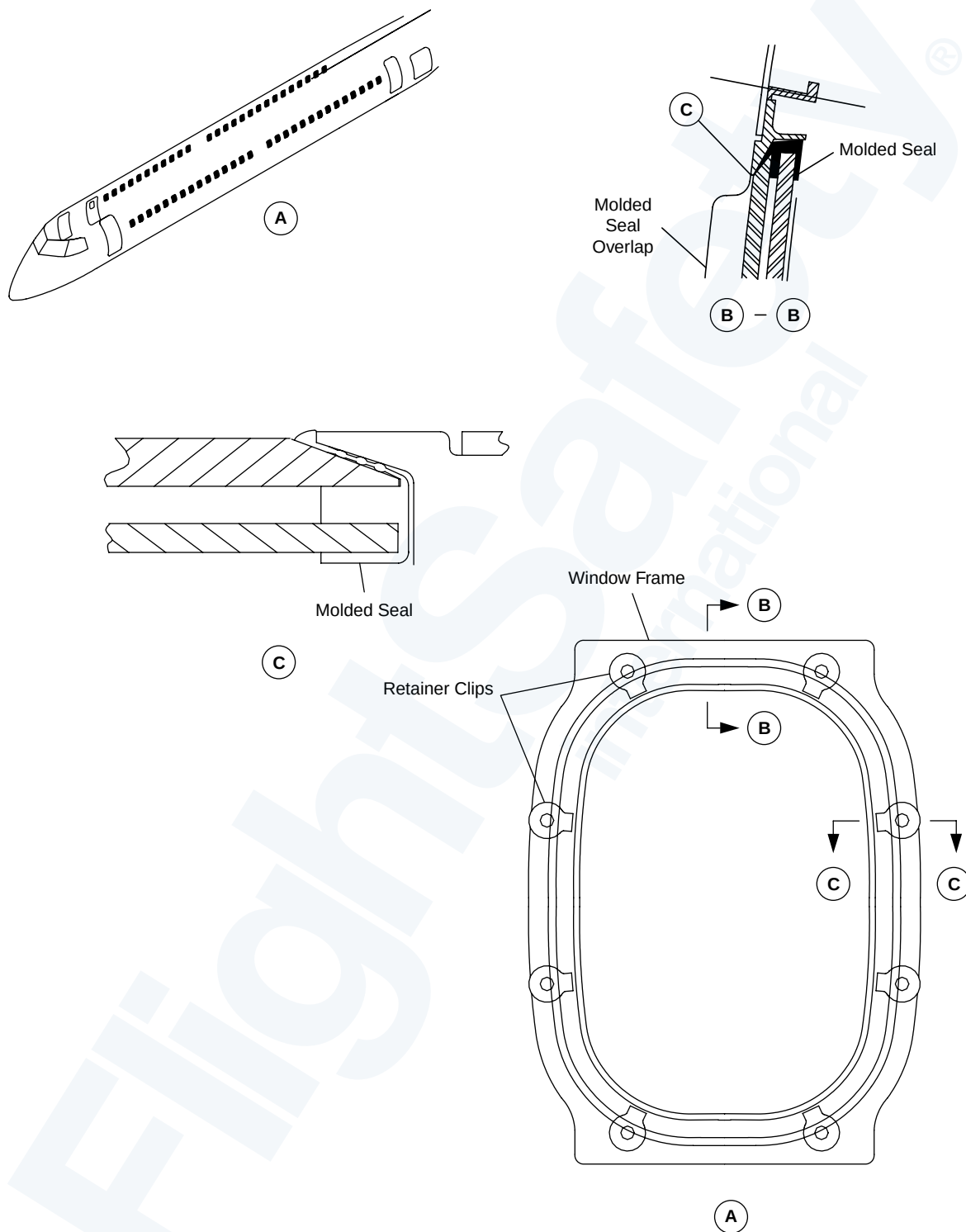
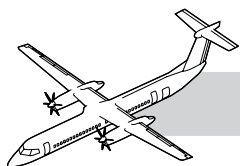
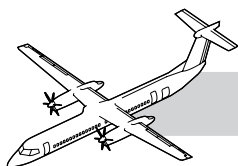


Figure 56-4. Passenger Compartment and Type II - Emergency-Exit-Door Windows



PASSENGER COMPARTMENT WINDOWS

Refer to:

- Figure 56-4. Passenger Compartment and Type II - Emergency-Exit-Door Windows
- Figure 56-5. Aft Passenger Door Window.

Windows are installed in each side of the passenger compartment and the doors that follow:

- Type II emergency-exit-door
- Aft passenger
- Service door.

Each window has an inner and outer acrylic pane in a molded silicone rubber seal.

The assembly is installed in an aluminium alloy window frame with retainer clips that are bolted to the frame.

The space between the panes of the passenger compartment windows are vented to the passenger compartment with two holes in the inner pane.

SPECIAL TOOLING

Commercially Available Wrench, Torque 0 to 40 lbf in (0 to 4.52 Nm).

SAFETY PRECAUTIONS

When you are near the windshield or side window, be very careful not to touch them with hard objects (tools, belt buckles, watches, jewellery). Hard objects can easily cause scratches or other damage to the windshield or side window.

When you remove or install the windshield or side window, make sure that you always put on cotton gloves. If you do not use cotton gloves you can cause scratches or damage to the windshield or side window. Scratches or damage can cause structural failure of the windshield or side window.

Do not use too much force or tools to remove the windshield. If you do, you can bend the retainer or cause damage to the frame or windshield.

Do not use lever bars or equivalent tools to set the windshield in the correct position. If you do, you can cause damage to the windshield.

Two people are required to remove and install the windshield in the aircraft structure.

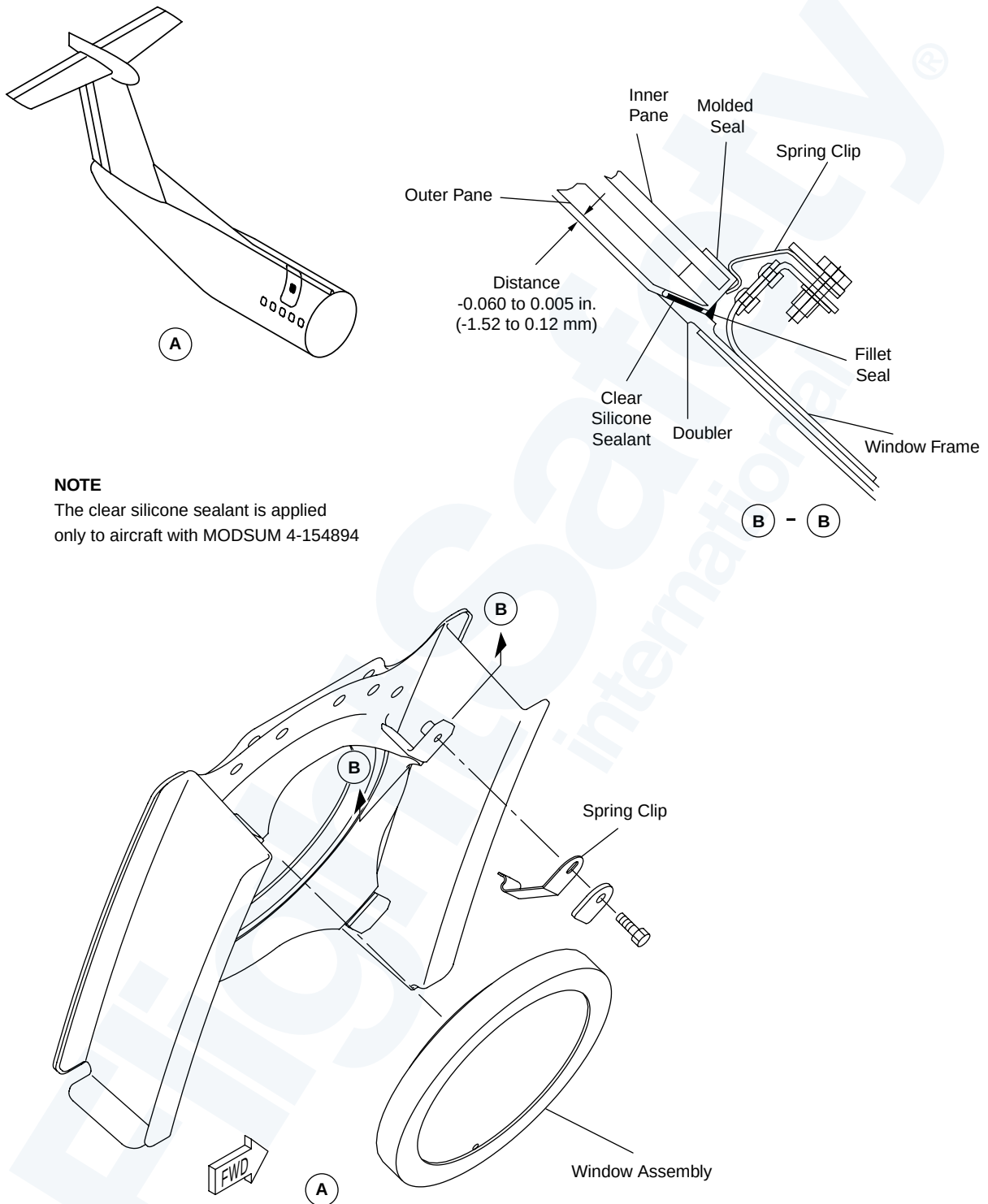
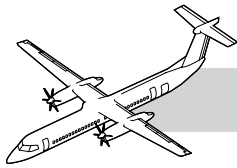


Figure 56-5. Aft Passenger Door Window